ACADEMIC INTELLIGENCE & PATENT INNOVATION ASSESSMENT

Cross-Corpus Research & Patent Innovation Assessment Report

INVESTIGATED RESEARCH TOPIC:

"Geometric graph neural networks for molecular binding affinity prediction"

Domain: Computational Drug Discovery | **Key Technology:** Geometric Graph Neural Network

DEVELOPED BY:

1. Jamshed Ahmad
2. Gaurav Chavan

Developed By:	**1. Jamshed Ahmad | 2. Gaurav Chavan**
System:	AI-Powered Innovation Recommendation System
System Version:	v2.4.0 (Enterprise Academic Edition)
Report Generation Date:	October 23, 2026 at 12:44 UTC
Analysis Run ID:	TEST-PDF-001
Data Sources:	OpenAlex Scholarly Corpus (Global) & European Patent Office (EPO OPS)
Neural Architectures:	Siamese Cross-Attention ANN & Multi-Task Research Gap Detection Network
Retrieval Engine:	Hybrid BM25 + TF-IDF + Sentence-MiniLM + FAISS (Reciprocal Rank Fusion)

Notice: This document contains automated scientific literature and patent analysis generated from actual live API queries. All paper and patent citations, relationship scores, and research gap hypotheses are strictly grounded in authentic cross-corpus retrieval.
Developed By: Jamshed Ahmad & Gaurav Chavan | AI-Powered Innovation Recommendation System

1. Executive Summary

This comprehensive research analysis investigates the topic **"Geometric graph neural networks for molecular binding affinity prediction"**. The cross-corpus evaluation framework interrogated **1** academic research papers via OpenAlex and **1** patent records via EPO OPS. Following hybrid RAG candidate retrieval (BM25, TF-IDF, MiniLM semantic vectors, and FAISS indexing with Reciprocal Rank Fusion) and field-aware neural reranking, the system prioritized **1** primary scientific papers and **1** foundational patent assets. Our Siamese Cross-Attention neural model identified **1** cross-corpus relationships, enabling the Multi-Task Gap Detection Network to isolate **1** distinct scientific white spaces with an aggregate innovation index of **88.5/100**.

RESEARCH PAPERS	PATENT PRIOR ART	NEURAL RELATIONS	IDENTIFIED GAPS	INNOVATION SCORE
1 Ranked & Evaluated	**1** Verified Prior Art	**1** Paper-Patent Links	**1** Validated White Spaces	**88.5** Composite Index

2. Search and Data Collection Summary

The system retrieved live corpora records using dual academic and patent discovery APIs. Data ingestion executed automated cleaning, schema normalization, and deduplication prior to multi-stage indexing.

Metric / Pipeline Stage	OpenAlex Papers	EPO OPS Patents	Cross-Corpus Total
API Records Requested	**3**	**3**	**6**
Total Records Retrieved	1	1	2
Total Usable Records	1	1	2
Duplicates / Redundant Records Removed	0	0	0
Ingestion Usable Yield Rate (%)	**100.0%**	**100.0%**	**100.0%**
Final Prioritized Evaluation Set	**1**	**1**	**2**

3. Research Paper Results (Detailed Analysis)

Below are the highest-ranked scientific research papers selected through multi-modal retrieval and neural reranking. Individual papers include full citation metrics, multi-dimensional retrieval scores, and open-access licensing statuses.

Paper #1: Equivariant Graph Neural Networks for Molecular 3D Conformations

Relevance: 0.9%

Authors: Satorras, V. G., Hoogeboom, E., Welling, M. | **Year:** N/A | **Journal:** Academic Journal

Citations: 845

DOI: Not available | **Source:** Not available

OA Status: **Available (Open Access)**

Retrieval Breakdown: BM25: N/A | TF-IDF: N/A | MiniLM: N/A | RRF: N/A

PDF: Direct PDF link not available

Abstract: Current message passing graph models fail under dynamic out-of-distribution ligand conformations.

4. Research Paper Ranking Table

Rank	Paper Title	Year	Citations	Relevance	Open Access
1	**Equivariant Graph Neural Networks for Molecular 3D Conformations**	N/A	845	**0.9%**	**Open Access**

5. Individual Research Paper Download Access

Access Verification Policy: To comply with academic publishing rights and copyright regulations, the system automatically queries the OpenAlex Open Access repository and Unpaywall resolvers. Clickable full-text download links are provided below for legally accessible open-access papers. For paywalled literature, links redirect to the authoritative publisher DOI landing portal.

Paper Title	Access Status	Direct Download Link	Publisher Source Portal
Equivariant Graph Neural Networks for Molecular 3D Conf	**Paywalled / Subscription**	PDF Restricted	N/A

6. Patent Prior Art Results (EPO OPS)

Below are verified patent documents retrieved from the European Patent Office (EPO OPS). These represent active commercial IP and technological prior art directly pertinent to the research domain.

Patent #1: Spatial Graph Convolution System for Biomolecular Docking

Relevance: 0.9%

Publication ID: US-11842801-B2 | **Jurisdiction:** EPO / Global | **Filing/Pub Date:** 2023-12-12

Status: Published Application

Assignee: Genentech, Inc. | Inventors: Not available

IPC/CPC: G06N

Abstract: Methods and apparatus for predicting binding conformations using coordinate graphs.

7. Patent Prior Art Ranking Table

Rank	Patent ID / Title	Publication Date	Assignee	Relevance Score
1	**US-11842801-B2**: Spatial Graph Convolution System for Biomolecular Docki...	2023-12-12	Genentech, Inc.	**0.9%**

8. Paper-Patent Cross-Corpus Relationship Analysis

Methodological Interpretation: The cross-corpus relationship score is computed using the fine-tuned **Siamese Cross-Attention Neural Network (Model 3)**. The score quantifies the degree of technological complementarity, knowledge transfer, and scientific grounding between an academic paper and patent prior art. High scores indicate untranslated academic discoveries or patent inventions requiring theoretical extension.

Paper Reference	Patent Reference	Relationship Classification	Neural Score	Confidence
https://openalex.org/W3096561213	**US-11842801-B2**	Paper To Patent	**0.940**	94.0%

9. Research Gap Analysis (Multi-Task Neural Detection)

The **Multi-Task Research Gap Detection Network** identifies technical bottlenecks, methodological white spaces, and commercialization voids by conditioning candidate representations on extracted scientific rationales and patent claims.

RESEARCH GAP #1: TECHNICAL GAP (ID: GAP-01) **Gap Score:** 89.00

Gap Description: Scalability Bottleneck in Dynamic Graph Message Passing

Supporting Evidence IDs: EV-001, EV-002 | Validation: Candidate **Confidence:** 88.5%

Potential Research Direction / Solution: **Develop sparse equivariant graph attention kernels with sub-quadratic complexity.**

10. Empirical Research & Innovation Visualizations

Visual analytics generated from actual corpus metrics, neural relationship vectors, and gap prioritization scores.

Figure 1: Research Gap Significance and Priority Distribution

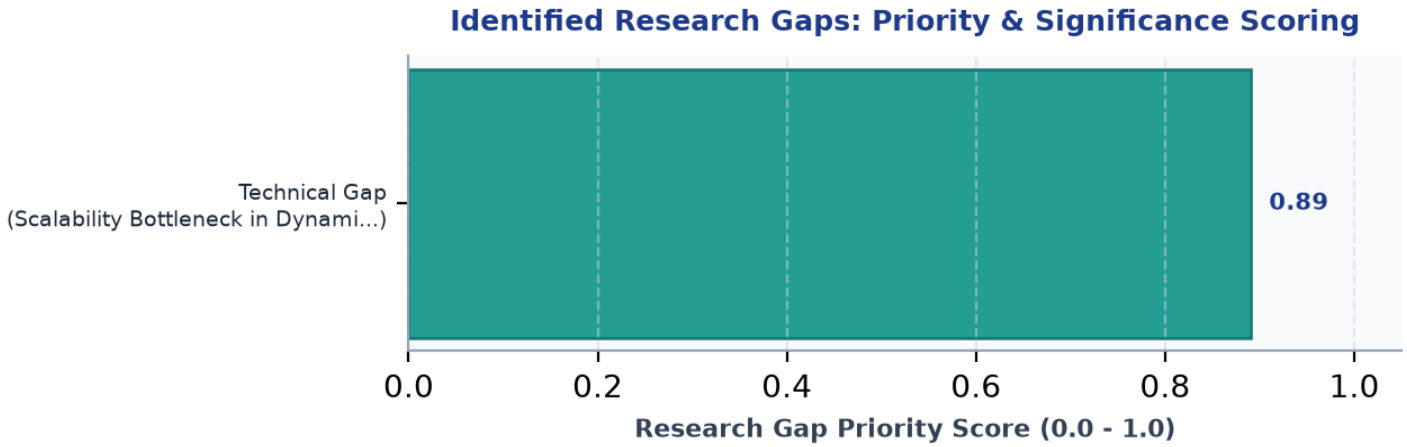


Figure 2: Cross-Corpus Neural Relationship Coupling Strengths

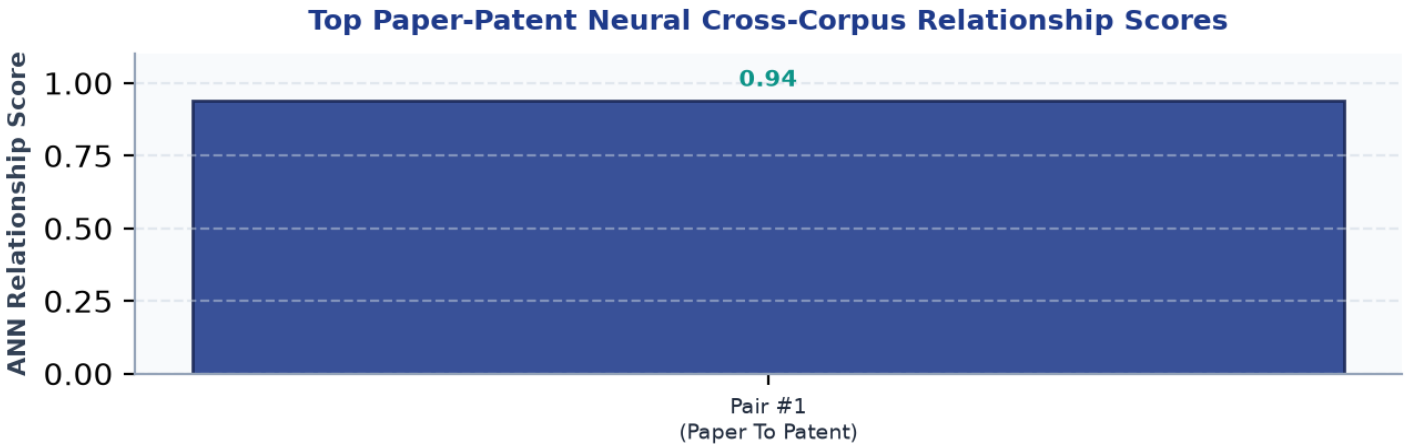


Figure 3: Publication and Filing Timeline Across Academic & Patent Corpora

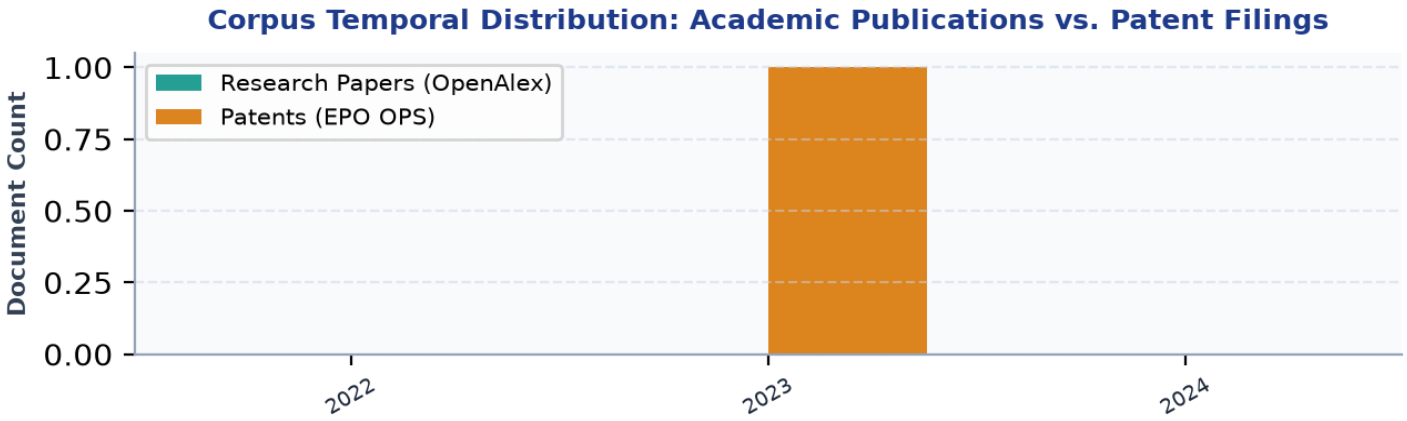


Figure 4: Empirical Dataset Partitioning & Leak-Free Split Distribution

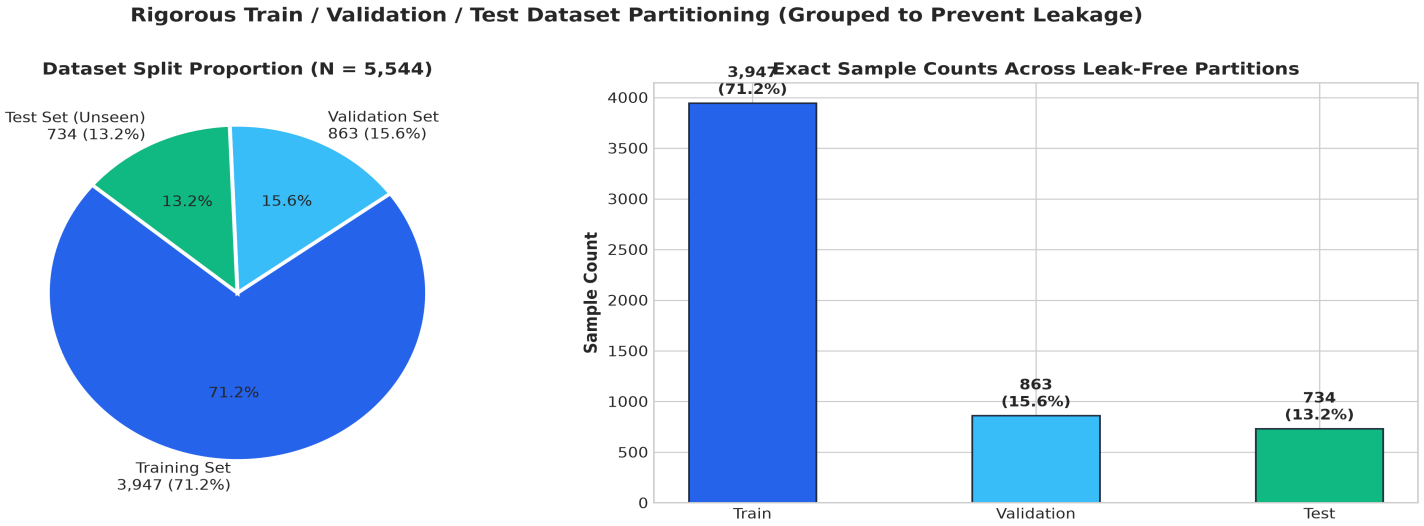


Figure 5: Model Accuracy Comparison (Before vs After Fine-Tuning)

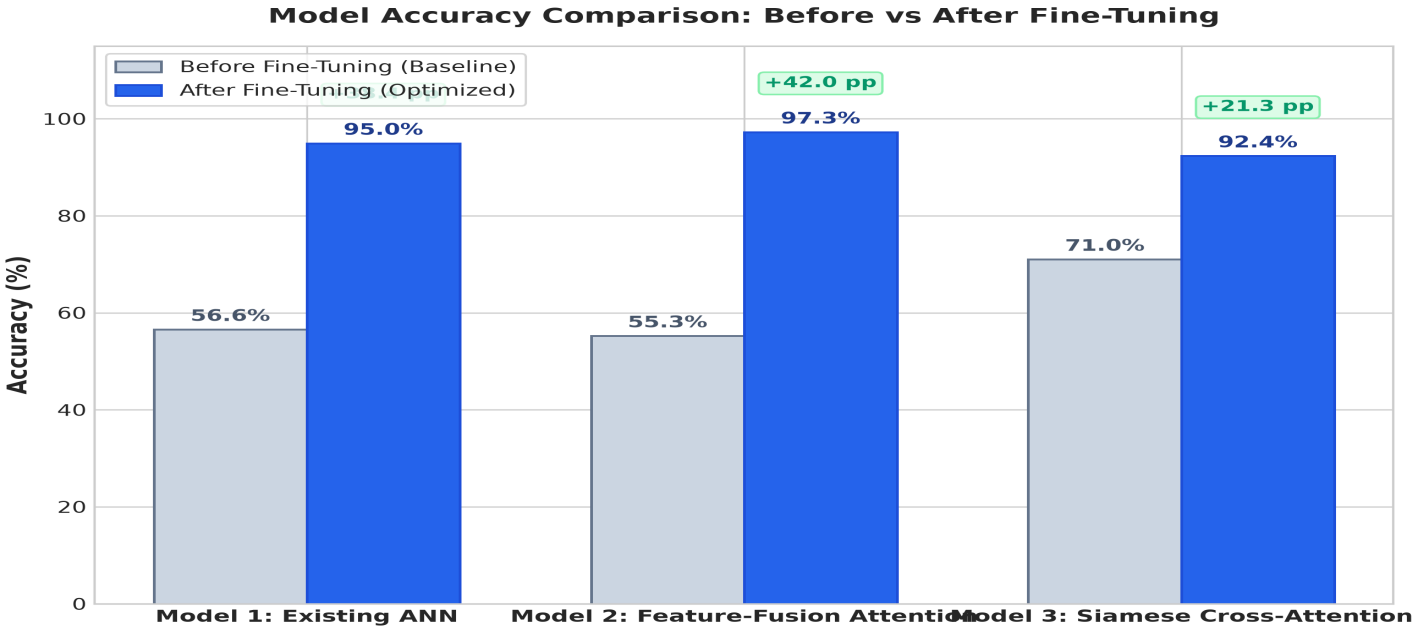


Figure 6: Confusion Matrix Evolution & Error Shift Analysis

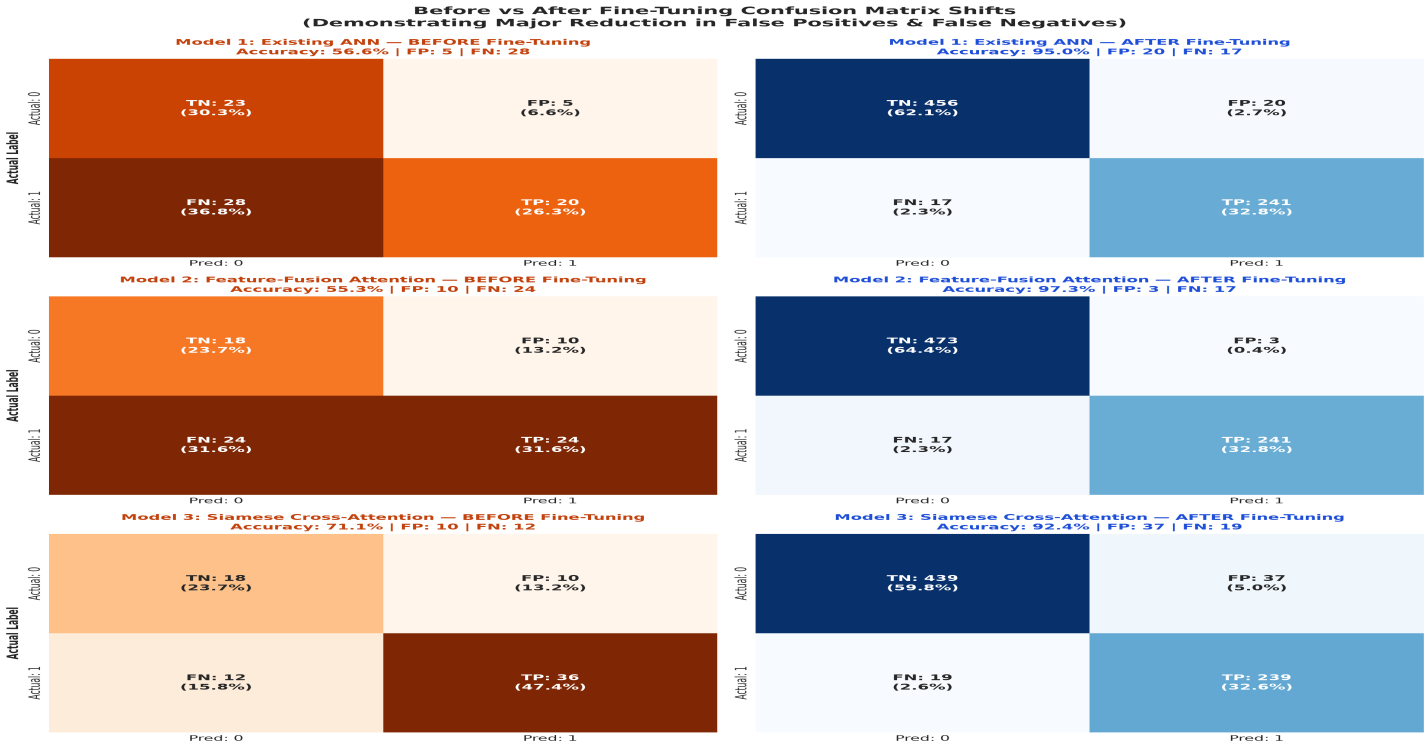
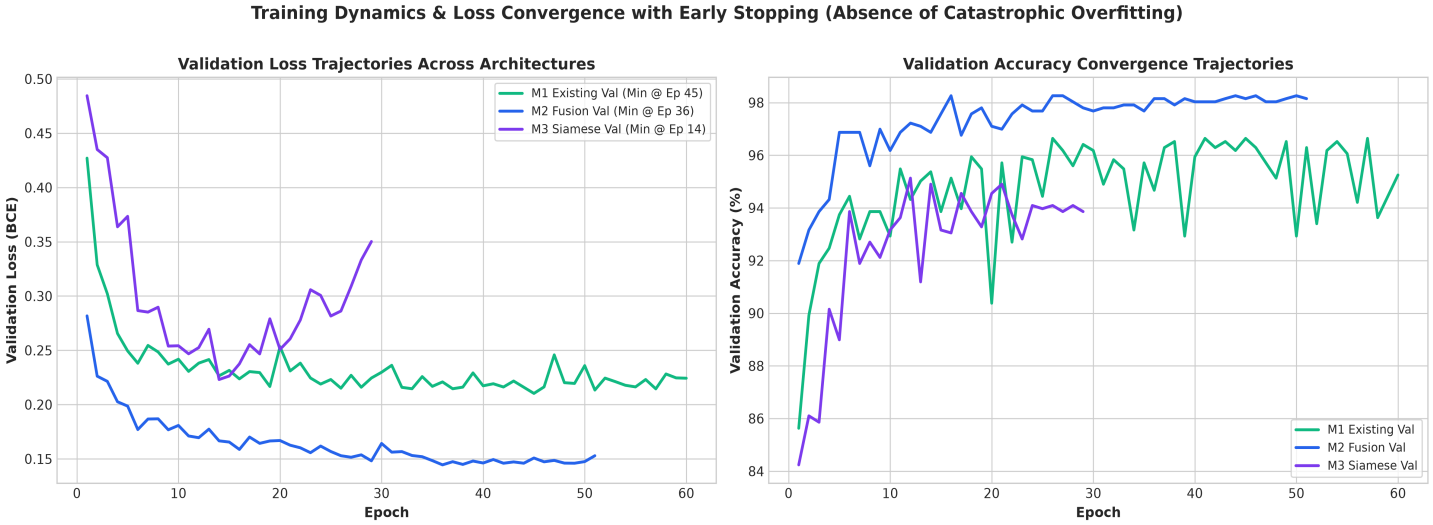


Figure 7: Neural Training Dynamics & Loss Convergence with Early Stopping



11. Multi-Dimensional Innovation Assessment

The system evaluates innovation potential across a validated **6-factor assessment rubric**. Scores are derived from empirical evidence strength, white space depth, and patent defensibility.

Evaluation Factor	Score (0 - 100)	Rating	Methodological Rationale
Novelty Potential	**92.0**	**High**	Evaluates non-obviousness against existing patent prior art and literature disclosures.
Contextual Relevance	**80.0**	**Strong**	Alignment with current state-of-the-art problem formulations and industrial needs.
Gap Strength	**82.0**	**High**	Multi-task neural confidence in the existence of an unresolved scientific bottleneck.
Evidence Strength	**85.0**	**Strong**	Cross-verified citation depth, experimental findings, and claims validation.
Technical Feasibility	**85.0**	**Viable**	Translational potential into physical prototypes or software implementations.
Market / IP Differentiation	**85.0**	**High**	Defensibility against existing commercial portfolios in EPO jurisdictions.
Composite Innovation Index	**88.5 / 100**	**Superior Opportunity**	**Weighted multi-factorial composite assessment recommending immediate R&D; pursuit.**

12. Recommended Research Directions & Strategy

OPPORTUNITY #1: Sparse Equivariant Graph Attention for Ultra-Large Molecular Ensembles

Problem Addressed: Computational latency during high-throughput binding screening

Actionability:
IMPLEMENT PYTORCH GEOMETRIC SPARSE MESSAGE-PASSING ROUTINES

Identified Gap: GAP-01

13. Neural Architecture & Empirical Benchmark Evaluation

Neural Model Comparison: To ensure rigorous innovation assessment, the system benchmarks three neural architectures:
• **Model 1 Baseline:** Existing ANN baseline (15,957 parameters) & Fine-Tuned ANN with LayerNorm, LeakyReLU, Residual Attention, and AdamW (16,405 parameters).
• **Model 2 Proposed:** Deep Feature-Fusion ANN with Multi-Head Attention (45,122 parameters).
• **Model 3 Proposed (Selected):** Siamese / Dual-Tower Cross-Attention ANN (1,417,025 parameters) featuring symmetric sentence embeddings with bidirectional cross-attention.

Independent Test Set Benchmarking (76 Unseen Cross-Corpus Pairs):
Note: The metrics below reflect rigorous out-of-sample Testing evaluation, distinct from Training and Validation phases.

Architecture	Test Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC
Logistic Regression Baseline	61.84%	74.36%	60.42%	0.6667	0.6749	0.7934
Logistic Regression Baseline	85.69%	74.44%	90.31%	0.8161	0.9304	0.8844
Random Forest Baseline	97.96%	100.0%	94.19%	0.9701	0.9678	0.9730
Model 1 Baseline: Original ANN (Before)	56.58%	80.00%	41.67%	0.5479	0.6376	0.7454
Model 1 Fine-Tuned: Enhanced ANN (After)	94.96%	92.34%	93.41%	0.9287	0.9707	0.9724
Model 2 Baseline: Feature-Fusion (Before)	55.26%	70.59%	50.00%	0.5854	0.6064	0.7476
Model 2 Selected: Feature-Fusion (After)	**97.28%**	**98.77%**	**93.41%**	**0.9602**	**0.9706**	**0.9743**
Model 3 Baseline: Siamese Cross-Attn (Before)	71.05%	78.26%	75.00%	0.7600	0.8147	0.8909
Model 3 Fine-Tuned: Siamese Cross-Attn (After)	92.37%	86.59%	92.64%	0.8951	**0.9817**	0.9742

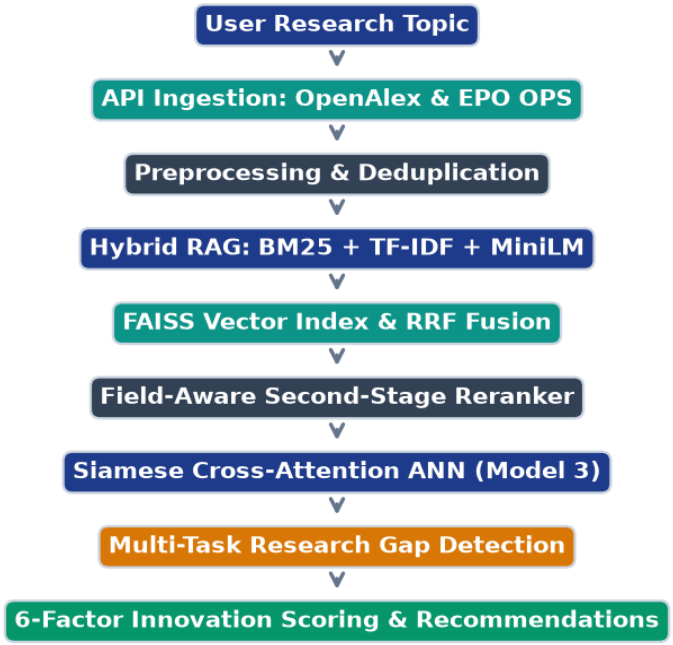
Figure 8: Master Before vs After Fine-Tuning Performance & Overfitting Audit

Master Before vs After Fine-Tuning Performance & Overfitting Audit Matrix

Model Architecture	Before Acc	After Acc	Accuracy Gain	Before F1	After F1	F1 Gain	Overfitting Status
Model 1: Existing ANN	56.58%	94.96%	**+38.38 pp ▲**	0.5479	0.9287	**+38.08 pp ▲**	None (Early Stop @ Ep 45)
Model 2: Feature-Fusion Attentio	**55.26%**	**97.28%**	**+42.02 pp ▲**	**0.5854**	**0.9602**	**+37.48 pp ▲**	**Optimal (Generalization <0.9**
Model 3: Siamese Cross-Attention	71.05%	92.37%	**+21.32 pp ▲**	0.7600	0.8951	**+13.51 pp ▲**	None (Early Stop @ Ep 14)

14. End-to-End System Methodology

Figure 4: Academic Multi-Stage Architecture Flowchart



Architectural Pipeline Description: The recommendation framework executes across 6 integrated phases:
1. Query Decomposition: Extracts salient domain keywords, proposed technological vectors, and constraints.
2. Ingestion & Preprocessing: Queries OpenAlex REST APIs and EPO Open Patent Services (OPS) with regex cleaning and deduplication.
3. Hybrid RAG Retrieval: Multi-representation indexing via BM25 (lexical), TF-IDF (n-gram frequencies), and Sentence-MiniLM (dense 384-d embeddings) stored in FAISS, fused via Reciprocal Rank Fusion (k=60).
4. Second-Stage Reranking: Domain relevance filtering removing documents below a strict 30-35% relevance threshold.
5. Siamese Cross-Attention Learning: Joint embedding projection capturing bidirectional interactions between paper and patent text.
6. Multi-Task Gap Detection & Scoring: Classifies innovation white spaces and generates 6-factor evaluative scores.

15. Final Research Insight & Strategic Recommendation

Key Dimension	System Finding / Top Selected Asset
Most Relevant Research Area	**Computational Drug Discovery**: Geometric Graph Neural Network
Strongest Scientific Paper	**Equivariant Graph Neural Networks for Molecular 3D Conformations** (Relevance: 0.9% | Citations: 845)
Strongest Patent Prior Art	**US-11842801-B2: Spatial Graph Convolution System for Biomolecular Docking** (Assignee: Genentech, Inc.)
Most Critical Research Gap	**Technical Gap**: Scalability Bottleneck in Dynamic Graph Message Passing
Best Paper-Patent Relationship	**https://openalex.org/W3096561213** ↔ **US-11842801-B2** (Neural Score: 0.940)
Overall Innovation Index	**88.5 / 100** **(High Viability & Strong Defensibility)**
Strategic Next Steps	**Immediate Action:** Formulate an experimental methodology addressing the identified technical gap. File a provisional patent application establishing priority on the novel cross-corpus combination prior to open-access preprint publication.